

Thesis Talk

Salome Bachmann

May 4, 2026



Overview of Topics

- Context
- Terminology
- Overview of Models
- Results of Cohesive Zone Model
- Discussion

What am I doing and why is this relevant?

- Hang out in my office and have coffee (please come by)
- Trying to make an accurate model for crack propagation in snow
- Cracks in snow on large scales can lead to avalanche release
- Average of 24 deaths by avalanche in Switzerland per year, multiple hundred non-fatal accidents (SLF, 2025)
- Understanding relevant because avalanches endanger human life and infrastructure (Schweizer et al., 2003)
- **Ultimate Goal:** determining whether slab breakoff can be seen in DAS data and how this would look

Words I will probably say a lot explained I

- Slab: Part of the snowpack which breaks off and slides downhill as a cohesive unit (the avalanche itself, I'm looking at dry-snow slab avalanches) (Schweizer et al., 2003)
- Weak layer: Layer with different mechanical properties than the rest of the snow, cracks usually propagate in this layer because it's weaker than the rest of the snowpack (Schweizer et al., 2003)
- Mode I, Mode II and Mode III cracks: cracks opening as a response to stress in different directions:

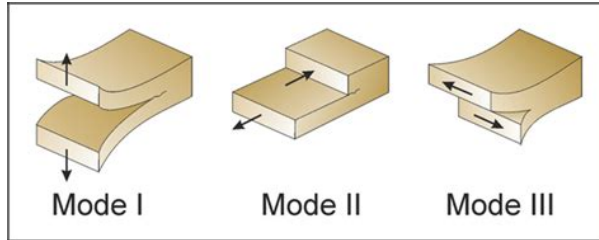


Figure: Fracture modes according to Philipp et al., 2013

Words I will probably say a lot explained II

- Anticrack: weak layer collapses under loading, slab starts bending, collapse and cracks caused by bending propagate (Gaume et al., 2018)
 - Sub-Rayleigh: Cracks that propagate at speeds lower than the shear wave speed in snow
 - Supershear: Cracks propagate faster than shear wave speed in snow

State of the art

- Two camps: Crack opening vs anticrack

Bibliography I



Gaume, J., Gast, T., Teran, J., van Herwijnen, A., & Jiang, C. (2018). Dynamic anticrack propagation in snow. *Nature Communications*, 9(1), 3047. <https://doi.org/10.1038/s41467-018-05181-w>



Philipp, S., Afşar, F., & Gudmundsson, A. (2013). Effects of mechanical layering on hydrofracture emplacement and fluid transport in reservoirs. *Frontiers in Earth Science*, 1. <https://doi.org/10.3389/feart.2013.00004>



Schweizer, J., Bruce Jamieson, J., & Schneebeli, M. (2003). Snow avalanche formation. *Reviews of Geophysics*, 41(4). <https://doi.org/10.1029/2002RG000123>



SLF. (2025). Avalanche Victims since 1936 - Long Term Statistics.



Salome Bachmann
Master Student
sbachmann@student.ethz.ch

ETH Zurich
D-EAPS